5.10 Complex Analysis

5.10.1 Cauchy-Riemann Equations

Proposition 5.10.1 (Isometry between CC and RR^2)
The canonical map $\chi : \mathbb{C} \to \mathbb{R}^2$ is a Banach space isometry (using the square norm $\|\cdot\|_2$). Furthermore define

$$\begin{aligned}\mathrm{Re}(z) &:= \pi_1\chi(z)\\ \mathrm{Im}(z) &:= \pi_2\chi(z)\end{aligned}$$

then

$$z = \mathrm{Re}(z)\cdot 1 + \mathrm{Im}(z)\cdot i$$

where $\mathbb{C}$ is viewed as an $\mathbb{R}$ vector space. One may also verify that complex multiplication may be represented in matrix form

$$\chi(z\cdot w) = \begin{pmatrix}\mathrm{Re}(z) & -\mathrm{Im}(z)\\ \mathrm{Im}(z) & \mathrm{Re}(z)\end{pmatrix}\chi(w)^T$$

Proposition 5.10.2 (Cauchy-Riemann Equations)
Let $U \subset \mathbb{C}$ and $f : U \to \mathbb{C}$ a function. There exists unique functions $u, v : \chi(U) \to \mathbb{R}$ such that

$$f(x+yi) = u(x,y) + v(x,y)i$$

Further the following are equivalent

a) $f(z)$ is continuously differentiable (as a complex function)
b) u, v are C^1 and satisfy the relations

$$\begin{aligned}\frac{\partial u}{\partial x}(x,y) &= \frac{\partial v}{\partial y}(x,y)\\ \frac{\partial u}{\partial y}(x,y) &= -\frac{\partial v}{\partial x}(x,y)\end{aligned}$$

Proof. We may define $F : \chi(U) \to \mathbb{R}^2$ by $F = \chi \circ f \circ \chi^{-1}$ and $u = \pi_1 \circ F$ and $v = \pi_2 \circ F$.

a) $\implies$ b) Fix $z = x + yi \in U$ and $f'(z) = a + bi$. Then for sufficiently small $w = h + ki$

$$f(z+w) - f(z) - f'(z)w$$

is sublinear in w. By assumption F is C^1 and has Jacobian matrix (...) at (x,y) equal to

$$J_F(x,y) := \begin{pmatrix}\frac{\partial u}{\partial x}(x,y) & \frac{\partial u}{\partial y}(x,y)\\ \frac{\partial v}{\partial x}(x,y) & \frac{\partial v}{\partial y}(x,y)\end{pmatrix}$$

and $Df(x,y)(h,k) = J_F(x,y)\begin{pmatrix}h\\k\end{pmatrix}^T$. However by conjugating the above equation by χ we find that

$$F(x+h, y+k) - F(x,y) - \begin{pmatrix}a & -b\\ b & a\end{pmatrix}\begin{pmatrix}h\\k\end{pmatrix}^T$$

(where $f'(z) = a + bi$) is sublinear (and using the fact χ is an isometry). Therefore the result then follows from the uniqueness of the total derivative $Df(x,y)$, and hence Jacobian J_F.

b) $\implies$ a) Conversely if the conditions hold then we may define $f'(z) := \frac{\partial u}{\partial x}(\chi(z)) + \frac{\partial v}{\partial x}(\chi(z))i$ which is evidently continuous. Then by using matrix representation of complex multiplication (5.10.1) we find

$$\chi[f(z+w) - f(z) - f'(z)w] = F(\chi(z+w)) - F(\chi(z)) - J'(\chi(z))\chi(w)^T$$

where

$$J'(x,y) = \begin{pmatrix}\frac{\partial u}{\partial x}(x,y) & -\frac{\partial v}{\partial x}(x,y)\\ \frac{\partial v}{\partial x}(x,y) & \frac{\partial u}{\partial x}(x,y)\end{pmatrix}$$

The conditions mean precisely that $J'(x,y) = J(x,y)$. As χ is an isometry we find that $f(z+w) - f(z) - f'(z)w$ is sublinear in w as required. ∎

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5.10.2 Holomorphic Functions

Definition 5.10.3 (Holomorphic Function)
*Let $U \subset \mathbb{C}$ be an open subset and $f : U \to \mathbb{C}$ a function. Then we say that f is **holomorphic** if it is infinitely (complex) differentiable (5.10.2). For a holomorphic function every derivative is automatically continuous (5.5.3).*

*If f is injective and $f(U)$ is open we say that f is **biholomorphic** if its inverse is also holomorphic. A biholomorphic functions is a homeomorphism onto its image (5.5.3).*

Trivially if f is holomorphic then so is $f|_V$ for any open subset $V \subset U$. Conversely if $U = \bigcup_{\alpha\in\mathcal{A}} V_\alpha$ and $f|_{V_\alpha}$ are holomorphic for all $\alpha \in \mathcal{A}$, then so is f.

Proposition 5.10.4
Let $U \subset \mathbb{C}$ be an open subset. Then the set of holomorphic functions on U forms a $\mathbb{C}$-algebra.

For $f : U \to \mathbb{C}$ non-constant then $D(f) := \{x \in U \mid f(x) \neq 0\}$ is an open subset and $\left(f|_{D(f)}\right)^{-1}$ is a holomorphic function.

Proof. This follows from (5.5.4), provided we show that $D(f)$ is open. For $\mathbb{C}$ is Hausdorff (5.4.5) so $\{0\}$ is closed (4.1.48). Therefore by (4.1.27) $D(f) = U \setminus f^{-1}(\{0\})$ is open. ∎

5.10.3 Exponential and Trigonometric Functions

Proposition 5.10.5 (Exponential Function)
Let K be a complete valued field. Then the series

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

has infinite radius of convergence. Furthermore it satisfies the following properties

a) e^z is infinitely differentiable with derivative e^z
b) $e^{z+w} = e^z e^w$
c) $e^z e^{-z} = 1$

In the case $K = \mathbb{R}$ then we also have the following property

d) e^x is positive, strictly increasing and bijective : $(-\infty, \infty) \to (0, \infty)$

Proof. We have

$$\left|\frac{a_{n+1}z^{n+1}}{a_n z^n}\right| = \frac{z}{n+1} \to 0$$

so the series converges unconditionally. Therefore by (5.6.2) it has infinite radius of convergence and is infinitely differentiable. Observe formally the product of power series is given by

$$\begin{aligned} e^z e^w &= \left(\sum_{n=0}^{\infty} \frac{z^n}{n!}\right)\left(\sum_{n=0}^{\infty} \frac{w^n}{n!}\right) \\ &= \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} \frac{z^k w^{n-k}}{k!(n-k)!}\right) \\ &= \sum_{n=0}^{\infty} \frac{(z+w)^n}{n!} \\ &= e^{z+w} \end{aligned}$$

which by (5.4.33) converge to the same value.

Evidently $x \geq 0 \implies e^x \geq 1 + x$. Then $x > y \geq 0 \implies e^x = e^{(x-y)+y} = e^{x-y}e^y \geq (1 + x - y)e^y > 1 \cdot e^y$. Similarly suppose $x < y \leq 0$ then $-y > -x \geq 0 \implies e^{-y} > e^{-x} \implies e^x < e^y$. Finally $x \leq 0 \implies e^{-x} \geq 1 - x \implies e^x \leq \frac{1}{1+x}$. This shows that e^x is strictly increasing and therefore injective on $(0, \infty)$. Further using $e^x e^{-x} = 1$ we see it is a positive function and $x \leq 0 \implies e^x \leq 1$.

As e^x is unbounded the intermediate value theorem (5.7.10) shows it achieves every value in $[1, \infty)$. Using $e^x e^{-x} = 1$ we see it also achieves every value in $(0, \infty)$. Therefore it is a bijection $(-\infty, \infty) \to (0, \infty)$ as required. ∎

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Proposition 5.10.6 (Trigonometric Functions)
The power series

$$\sin(z) = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots$$

$$\cos(z) = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots$$

have infinite radius of convergence in $\mathbb{C}$*. Furthermore*

a) $\sin(0) = 0$ *and* $\cos(0) = 1$
b) $e^{iz} = \cos(z) + i\sin(z)$
c) $\sin(z) = \frac{e^{iz} - e^{-iz}}{2i}$ *and* $\cos(z) = \frac{e^{iz} + e^{-iz}}{2}$*.* $\sin(-z) = -\sin(z)$ *and* $\cos(-z) = \cos(z)$
d) $\sin'(z) = \cos(z)$ *and* $\cos'(z) = -\sin(z)$ *are infinitely differentiable*
e) $\sin^2(z) + \cos^2(z) = 1$
f) $\sin(w + z) = \sin(w)\cos(z) + \cos(w)\sin(z)$
g) $\cos(w + z) = \cos(w)\cos(z) - \sin(w)\sin(z)$

Proof. It is simpler to define $\sin(z)$ and $\cos(z)$ by *c)*. The power series expansion then follows from (5.4.26).

a) Immediate from the series expansion
b) Immediate from *c)*
c) Already shown
d) Immediate from *c)* and the chain rule (5.9.6).
e) Use *b)* and $e^{iz}e^{-iz} = 1$

f)-g) Using *b)* we find

$$\begin{aligned}\cos(z + w) + i\sin(z + w) &= e^{i(z+w)} = e^{iz}e^{iw} \\ &= (\cos(z) + i\sin(z))(\cos(w) + i\sin(w)) \\ &= (\cos(z)\cos(w) - \sin(z)\sin(w)) + i(\sin(z)\cos(w) + \cos(z)\sin(w))\end{aligned}$$

and the result follows from equating real and imaginary parts ∎

Proposition 5.10.7 (cos and sin are periodic)
There is a unique positive real number $\pi \in \mathbb{R}$ *such that the real roots of* $\sin(x)$ *are precisely*

$$\{n\pi \mid n \in \mathbb{Z}\}$$

Furthermore we have the period formulae

a) $\sin(x) = \cos(x - \frac{\pi}{2}) = \cos(\frac{\pi}{2} - x)$
b) $\cos(x) = \sin(\frac{\pi}{2} - x) = -\sin(x - \frac{\pi}{2})$
c) $\sin(x) = -\sin(x + \pi)$ *and* $\sin(x + 2\pi) = \sin(x)$
d) $\cos(x) = -\cos(x + \pi)$ *and* $\cos(x + 2\pi) = \cos(x)$

and the real roots of $\cos(x)$ *are precisely*

$$\left\{\left(n + \frac{1}{2}\right)\pi \;\middle|\; n \in \mathbb{Z}\right\}$$

Proof. By (5.10.6).e) we know that $0 \leq \sin^2(x) \leq 1$ which means $-1 \leq \sin(x) \leq 1$, and similarly for $\cos(x)$. We first prove that $\cos(x)$ has a positive root. Suppose not then by the intermediate value theorem (5.7.10) we must have $\cos(x) > 0$ for all x. Then by the Mean-Value Theorem (5.7.12) $\sin(x)$ must be strictly increasing and therefore positive for all $x > 0$. We claim that for $x \geq a > 0$

$$\cos(x) \leq \cos(a) + \cos'(a)(x - a) = \cos(a) - \sin(a)(x - a)$$

For define $g(x) \coloneqq \cos(a) - \sin(a)(x - a) - \cos(x)$ then $g(a) = \cos(a) > 0$ and $g'(x) = \sin(x) - \sin(a) > 0$. By the Mean-Value Theorem $g(x)$ is increasing and the claim is proven. However it is evident that this implies $\cos(x)$ is unbounded below contradicting $-1 \leq \cos(x)$. Therefore $\cos(x)$ has a strictly positive root.

Define this to be

$$\frac{\pi}{2} \coloneqq \inf\{x \in \mathbb{R}^+ \mid \cos(x) = 0\}$$

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By (5.1.18) there exists a sequence x_n such that $\cos(x_n) = 0$ and $x_n \uparrow \frac{\pi}{2}$. By continuity and (5.3.6) we have $\cos(\frac{\pi}{2}) = 0$, and evidently $\pi \neq 0$. We must have $\cos(x)$ positive on the interval $[0, \frac{\pi}{2}]$, for otherwise by the intermediate value theorem we would have a smaller root. This shows that $\sin(x)$ is increasing on the interval $[0, \frac{\pi}{2}]$. By (5.10.6) we have $\sin^2(\frac{\pi}{2}) = 1$ and therefore $\sin(\frac{\pi}{2}) = 1$.

The period formula $a)$ then follows from the addition formula (5.10.6).g)

$$\cos\Big(\frac{\pi}{2} - x\Big) = \cos\Big(x - \frac{\pi}{2}\Big) = \cos(x)\cos\Big(-\frac{\pi}{2}\Big) - \sin(x)\sin\Big(-\frac{\pi}{2}\Big) = \sin(x)$$

and $b)$ follows directly from $a)$. Then $c), d)$ follow from combining $a), b)$. In particular $\sin(n\pi) = 0$. We claim that π is the smallest positive root of $\sin(x)$. For suppose $0 < x < \pi$ satisfies $\sin(x) = 0$ then by $a)$ we have $\cos\big(\big|x - \frac{\pi}{2}\big|\big) = 0$ and $0 \leq \big|x - \frac{\pi}{2}\big| < \frac{\pi}{2}$ which contradicts the choice of π. Suppose $\sin(y) = 0$ with $y > 0$. Then consider

$$n := \max\{m \in \mathbb{N} \mid y - m\pi \geq 0\}$$

Then $\sin(y - n\pi) = 0$ by $c)$. By choice of m we have $0 \leq y - n\pi < \pi$. As π is the smallest positive root we have $y - n\pi = 0$. By symmetry then the roots of $\sin(x)$ are precisely $\{n\pi\}$. Evidently any such π is unique because $\pi' = n\pi = nm\pi'$ whence $n = m = 1$. Using $a)$ yields the precise set of roots of $\cos(x)$. ∎

Proposition 5.10.8
There is a continuous, strictly decreasing function

$$\arccos : [-1, 1] \to [0, \pi]$$

such that

$$\cos(\arccos(x)) = x$$

Proof. By (5.10.7) $\sin(x)$ is positive on the interval $(0, \pi)$ (by the intermediate value theorem). By the Mean Value Theorem then $\cos(x)$ is strictly decreasing on this interval. Furthermore $\cos(0) = 1$ and $\cos(\pi) = -\cos(\pi - \pi) = -1$. By the intermediate value theorem then $\cos : [0, \pi] \to [-1, 1]$ is surjective and injective. Therefore the map $\arccos$ is well-defined and it is continuous by (5.7.13). ∎

Proposition 5.10.9 (Polar Coordinates)
There is a unique function

$$\theta : \mathbb{R}^2 \setminus \{(0, 0)\} \to [0, 2\pi)$$

such that

$$r(x, y)\cos(\theta(x, y)) = x \text{ and } r(x, y)\sin(\theta(x, y)) = y$$

where $r(x, y) := \sqrt{x^2 + y^2}$. *Explicitly it is given by*

$$\theta(x, y) := \begin{cases} \arccos\big(\frac{x}{r(x,y)}\big) & y \geq 0 \\ 2\pi - \arccos\big(\frac{x}{r(x,y)}\big) & y < 0 \end{cases}$$

and it is continuous on $\mathbb{R}^2 \setminus \{(x, 0) \mid x > 0\}$, *and* $\theta(x, 0-) = 2\pi + \theta(x, 0+)$ *when* $x > 0$.

Proof. Observe that $0 \leq \frac{x}{\sqrt{x^2+y^2}} \leq 1$ and so $r(x, y)\cos(\theta(x, y)) = x$. Further $\sin^2(\theta(x, y)) = 1 - \cos^2(\theta(x, y)) = \frac{y^2}{x^2+y^2}$, and so $r(x, y)\sin(\theta(x, y)) = \pm y$. When $y \geq 0$ we have $\theta(x, y) \in [0, \pi] \implies \sin(\theta(x, y)) \geq 0$. Similarly from the case $y \leq 0$ we find $r(x, y)\sin(\theta(x, y)) = y$ as required. ∎

Proposition 5.10.10
Let $r, r' > 0$ *and* $\theta, \theta' \in \mathbb{R}$. *Then*

$$(r\cos(\theta), r\sin(\theta)) = (r'\cos(\theta'), r'\sin(\theta'))$$

if and only if

$$r = r' \text{ and } \theta = \theta' + 2n\pi$$

Proof. We may use (5.10.6) to show that

$$r^2 = (r\cos(\theta))^2 + (r\sin(\theta))^2 = (r'\cos(\theta'))^2 + (r'\sin(\theta'))^2 = (r')^2$$

which means $r = r'$. Similarly we may use the addition formula to find

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$$\sin(\theta - \theta') = \sin(\theta)\cos(\theta') - \cos(\theta')\sin(\theta) = 0$$

which means $\theta = \theta' + n\pi$ by (5.10.7). Using the fact $\cos(\theta + n\pi) = (-1)^n \cos(\theta)$ and $\sin(\theta + n\pi) = (-1)^n \sin(\theta)$ we may deduce that n is even (since $\sin(\theta) = \cos(\theta) = 0$ is impossible). ∎

5.10.4 Polar Coordinates of a Path

We would like to show that any path $\gamma : [a, b] \to \mathbb{C} \setminus \{0\}$ has a *continuous* parameterisation $(r(t), \theta(t))$ in polar coordinates, and that this is essentially unique.

Definition 5.10.11

*Suppose X is a Hausdorff space. A continuous surjective map $p : \tilde{X} \to X$ is a **covering map** if every $x \in X$ has a neighbourhood U_x such that*

a) $p^{-1}(U_x) = \bigsqcup_{i \in I_x} V_{x,i}$

b) $p|_{V_{x,i}} : V_{x,i} \to U_x$ is a homeomorphism for all $i \in I_x$

We may show that $\tilde{X}$ is necessarily Hausdorff.

Proposition 5.10.12 (Punctured Plane Covering Map)

The map

$$\begin{aligned} \mathbb{R}_{>0} \times \mathbb{R} &\to \mathbb{R}^2 \setminus \{(0,0)\} \\ (r, \theta) &\to (r\cos(\theta), r\sin(\theta)) \end{aligned}$$

is a covering map.

Proof. Define

$$\begin{aligned} U_1 &:= \mathbb{R}^2 \setminus \{(x, 0) \mid x \geq 0\} \\ U_2 &:= \mathbb{R}^2 \setminus \{(x, 0) \mid x \leq 0\} \end{aligned}$$

We claim that

$$\begin{aligned} p^{-1}(U_1) &= \bigsqcup_n \mathbb{R}_{>0} \times (2n\pi, (2n+2)\pi) \\ p^{-1}(U_2) &= \bigsqcup_n \mathbb{R}_{>0} \times ((2n-1)\pi, (2n+1)\pi) \end{aligned}$$

Denote the open sets by $V_{i,n}$ for $i = 1, 2$. Then the maps

$$p|_{V_{i,n}} \to U_i$$

are injective by (5.10.10). We may define explicit continuous inverses by

$$\begin{aligned} U_1 &\to V_{1,n} \\ (x, y) &\to \left(\sqrt{x^2 + y^2}, \theta(x, y) + 2n\pi\right) \\ U_2 &\to V_{2,n} \\ (x, y) &\to \left(\sqrt{x^2 + y^2}, -\theta(-x, y) + 2(n-1)\pi\right) \end{aligned}$$

∎

Proposition 5.10.13 (Path lifting)

Let $p : \tilde{X} \to X$ be a covering map and $\gamma : [a, b] \to X$ a continuous path. Consider any $\tilde{x}_0 \in \tilde{X}$ such that $p(\tilde{x}_0) = \gamma(a)$. Then there exists a unique lifting $\tilde{\gamma} : [a, b] \to \tilde{X}$ such that $p \circ \tilde{\gamma} = \gamma$ and $\tilde{\gamma}(a) = \tilde{x}_0$.

Proof. We first prove uniqueness. Consider two liftings $\tilde{\gamma}_1, \tilde{\gamma}_2 : [a, b] \to \tilde{X}$ and define

$$X := \{t \in [a, b] \mid \tilde{\gamma}_1(t) = \tilde{\gamma}_2(t)\}$$

Fix $t \in X$ and let U be a neighbourhood of $\gamma(t)$ such that $p^{-1}(U) = \bigsqcup_{i \in I} V_i$. Suppose $\tilde{\gamma}_1(t) = \tilde{\gamma}_2(t) \in V_i$. By continuity there exists a neighbourhood W of t such that $\tilde{\gamma}_1(W) \subseteq V_i$ and $\tilde{\gamma}_2(W) \subseteq V_i$. By definition of X this means $W \subseteq X$. Therefore X is open. By (4.1.99) X is also closed. By (5.7.9) then X is either empty or $[a, b]$. However we have $a \in X$ and the liftings must be equal.

Define

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$$L := \left\{ x \in [a,b] \;\middle|\; \exists \tilde{\gamma} : [a,x] \to \tilde{X} \text{ s.t. } p \circ \tilde{\gamma} = \gamma \text{ and } \tilde{\gamma}(a) = \tilde{x}_0 \right\}$$

To show that L is closed consider a sequence $x_n \in L$ such that $x_n \to x \in [a,b]$. Choose $\gamma(x) \in U$ such that $p^{-1}(U) = \bigsqcup_i V_i$ and $p|_{V_i}$ is a homeomorphism. For some ε we have $(x-\varepsilon, x+\varepsilon) \cap [a,b] \subset \gamma^{-1}(U)$, and for some N, $x_N \in (x-\varepsilon, x+\varepsilon) \cap [a,b]$. In particular there is a lift

$$\tilde{\gamma}' : [a, x_N] \to \tilde{X}.$$

and $\tilde{\gamma}'(x_N) \in V_i$ for some i. If $x_N \geq x$ then we are done, otherwise assume $x_N < x$ and define the lift

$$\tilde{\gamma}(t) := \begin{cases} \tilde{\gamma}'(t) & t \leq x_N \\ \left(p|_{V_i}\right)^{-1}(\gamma(t)) & x_N \leq t \leq x \end{cases}$$

Therefore $x \in L$, and L is closed. By (5.7.9) we conclude $L = [a,b]$ and we are done. ∎

Proposition 5.10.14 (Almost uniqueness of polar coordinates)
Let $p : \mathbb{R}_{>0} \times \mathbb{R} \to \mathbb{R}^2 \setminus \{(0,0)\}$ be the covering map of the punctured plane (5.10.12) and $\gamma : [a,b] \to \mathbb{R}^2 \setminus \{(0,0)\}$ a continuous path. Suppose (r_1, θ_1) and (r_2, θ_2) are liftings. Then

$$r_1(t) = r_2(t)$$

$$\theta_1(t) = \theta_2(t) + 2n\pi$$

for some $n \in \mathbb{Z}$.

Proof. By (5.10.10) we have $\theta_2(a) - \theta_1(a) = 2n\pi$. Then $(r_2(t), \theta_2(t) - 2n\pi)$ is also a lifting and so by uniqueness equals $(r_1(t), \theta_1(t))$ as required. ∎

Definition 5.10.15 (Winding Number)
*Let $\gamma : [a,b] \to \mathbb{C} \setminus \{0\}$ be a continuous path. We define the **angle traversed** by the path γ to be*

$$\theta(b) - \theta(a)$$

*where $(r(t), \theta(t))$ is any lifting of γ. Note when γ is a closed path then this equals $2n\pi$ for a unique integer n. We call this the **winding number** of the closed path γ, which we denote by $W(\gamma; 0)$. In general we define*

$$W(\gamma; z_0) := W(\gamma - z_0; 0)$$

for any $z_0 \notin \gamma([a,b])$.

5.10.5 Path Integrals

Definition 5.10.16
*A **curve** is a continuous path $\gamma : [a,b] \to \mathbb{C}$, for which there exists a partition*

$$a = x_0 < x_1 < \cdots < x_{n+1} = b$$

such that

a) γ is C^1 on (x_i, x_{i+1}) for $i = 0 \cdots n-1$
b) γ is left and right differentiable at x_i and

$$\lim_{h \downarrow 0} \frac{\gamma(x_i + h) - \gamma(x_i)}{h} = \lim_{h \downarrow 0} \gamma'(x_i + h)$$

$$\lim_{h \uparrow 0} \frac{\gamma(x_i + h) - \gamma(x_i)}{h} = \lim_{h \uparrow 0} \gamma'(x_i + h)$$

We say γ is closed if in addition $\gamma(b) = \gamma(a)$.

Definition 5.10.17 (Path Integral)
Let $\gamma : [a,b] \to \mathbb{C}$ be a curve supported on $\{x_0, \cdots, x_{n+1}\}$. Define the integral

$$\int_\gamma f(z)\,\mathrm{d}z := \sum_{i=0}^{n} \int_{x_i}^{x_{i+1}} f(\gamma(t))\gamma'(t)\,\mathrm{d}t$$

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Proposition 5.10.18 (Path Integral of a function with primitive)
Let $f : U \to \mathbb{C}$ be a continuous function and $g : U \to \mathbb{C}$ a differentiable function such that $g'(z) = f(z)$. Then for any curve $\gamma : [a, b] \to U$ we have

$$\int_\gamma f(z)\,\mathrm{d}z = g(b) - g(a)$$

In particular the integral over a closed curve is zero.

Proposition 5.10.19
Let $\gamma : [0, 1] \to \mathbb{C}$ the path given by $\gamma(t) \coloneqq Re^{2\pi i t}$. Then evidently $W(\gamma, 0) = 1$. Further

$$\int_\gamma z^n\,\mathrm{d}z = \begin{cases} 2\pi i & n = -1 \\ 0 & n \neq -1 \end{cases}$$

Proof. When $n \neq -1$ then z^n has primitive $\frac{z^{n+1}}{n+1}$ and we are done by the previous result. In the case $n = -1$ we may calculate explicitly

$$\int_\gamma z^{-1}\,\mathrm{d}z = \int_0^1 \frac{\gamma'(t)}{\gamma(t)}\,\mathrm{d}t = 2\pi i$$

■

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